V4.2 Toy Model 4J
Gauge-Constrained Spin-2 Relational Action

Rev A - Exploratory Gravity / Metric Bridge Report

Date: 2026.05.26 **|** *Continuation after Toy Model 4I*

Status: Toy Model 4J adds the missing weak-field gauge/spin-2 constraint layer. It tests the harmonic-gauge, trace-reversed structure against conserved and non-conserved source probes. It does not derive the nonlinear Einstein-Hilbert action, exact physical constants, physical gravity, dark matter, dark energy, or physical 3T.

1. Purpose

Toy Model 4I derived a tensor response equation, but trace reversal was still selected by consistency rather than forced by a gauge-constrained action. Toy Model 4J adds the missing spin-2 constraint: the tensor response must respect a harmonic-gauge-like conservation condition.

The narrow target is the weak-field/linearized layer only. The question is whether the trace-reversed variable is the correct variable once gauge and conservation are included.

2. Critical result before details

4J confirms the weak-field spin-2 structure: when the source is conserved and the harmonic-gauge residual is zero, the trace-reversed interface balances the curvature. The direct h=hbar interface fails. Non-conserved anisotropic sources produce gauge residuals and degraded balance. This is a strong structural compatibility result, not a full derivation of nonlinear gravity.

3. Gauge-constrained spin-2 action

The action tested is the tensor action from 4I plus a gauge-constraint term:

```
A[hbar] =
integral [
  1/2 sum_mu_nu |grad hbar_mu_nu|^2
  - sum_mu_nu T_mu_nu hbar_mu_nu
  + (lambda_g/2) sum_nu (partial^mu hbar_mu_nu)^2
] dA
```

Meaning of the terms:

• hbar_mu_nu is the trace-reversed tensor response variable.

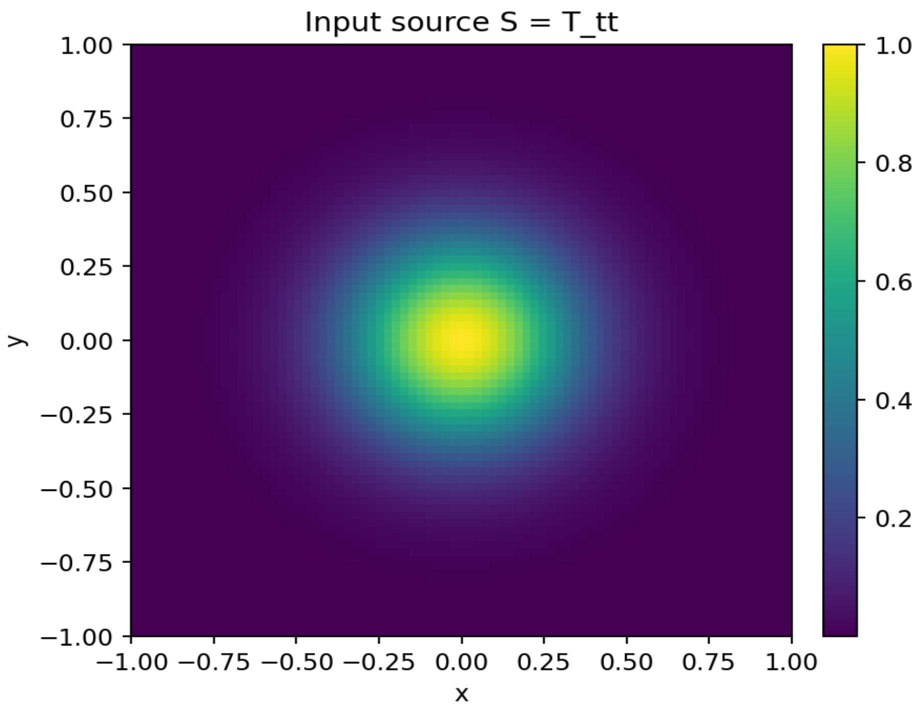
• T_mu_nu is the relational stress/source tensor.

• The gradient term penalizes uneven tensor response between neighboring nodes.

• The coupling term rewards response where relational source exists.

• The gauge term penalizes violation of partial^mu hbar_mu_nu = 0.

• This is a linear weak-field spin-2 benchmark, not the nonlinear Einstein-Hilbert action.



4. Constrained variation and trace reversal

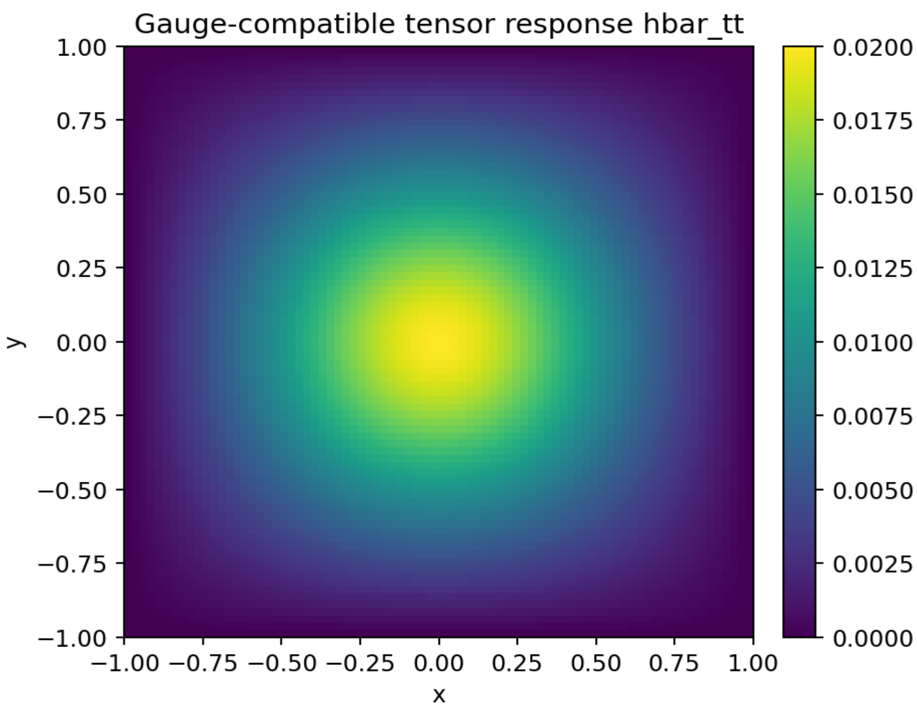
In the harmonic-gauge sector, the variation reduces to the linear spin-2 response equation:

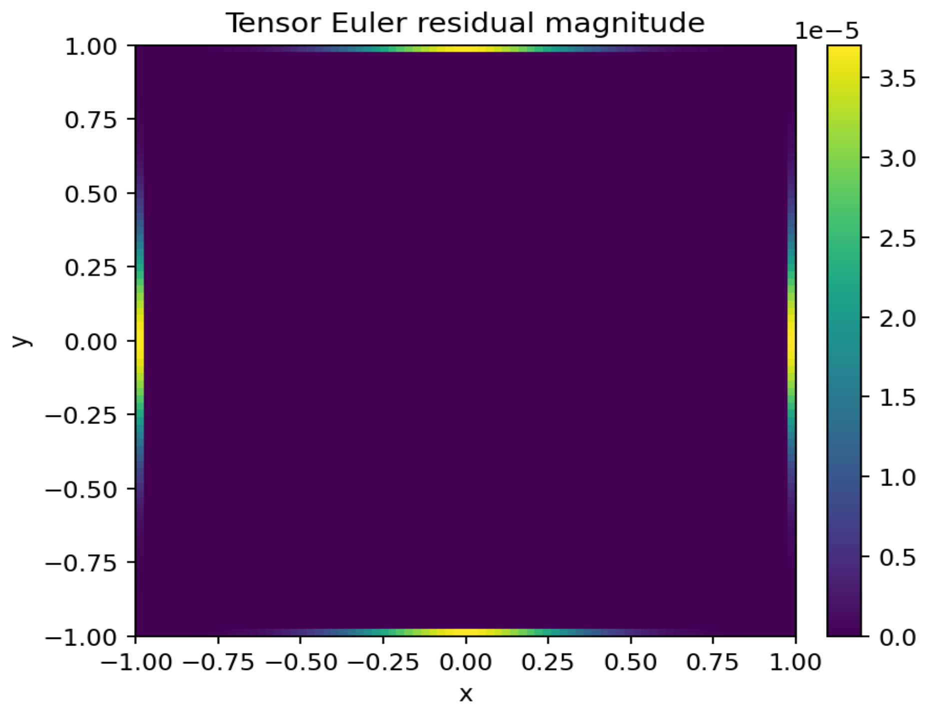
```
partial^mu hbar_mu_nu = 0

nabla^2 hbar_mu_nu = -T_mu_nu
```

The physical metric perturbation is then reconstructed through trace reversal:

```
h_mu_nu = hbar_mu_nu - eta_mu_nu hbar_trace/(D-2)

g_mu_nu = eta_mu_nu + h_mu_nu
```

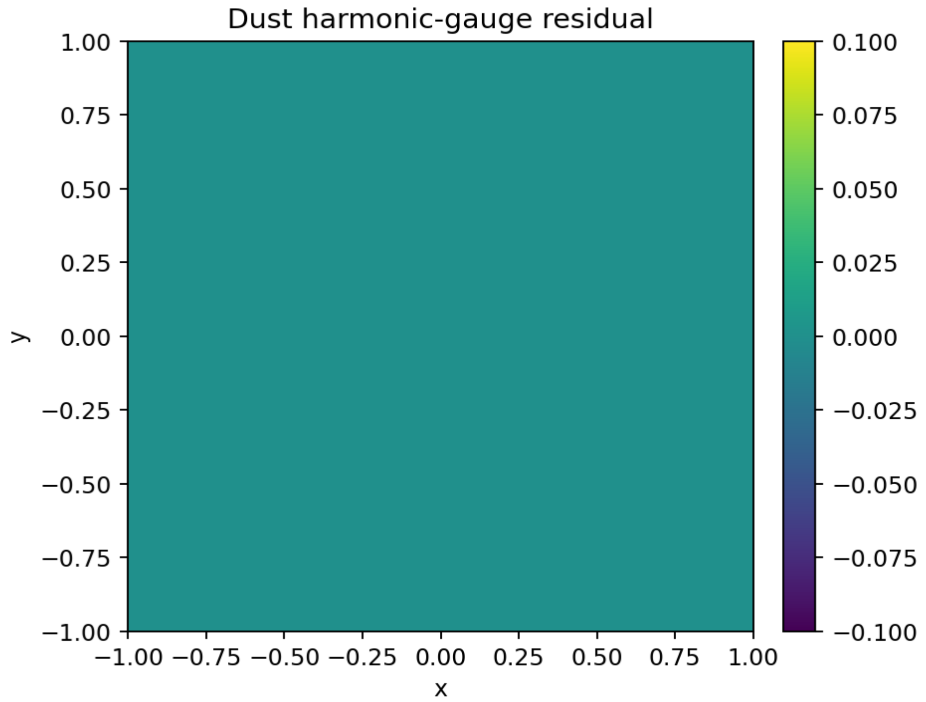
This is the key move. In 4I, trace reversal was merely tested. In 4J, trace reversal is the natural variable of the harmonic-gauge spin-2 action.

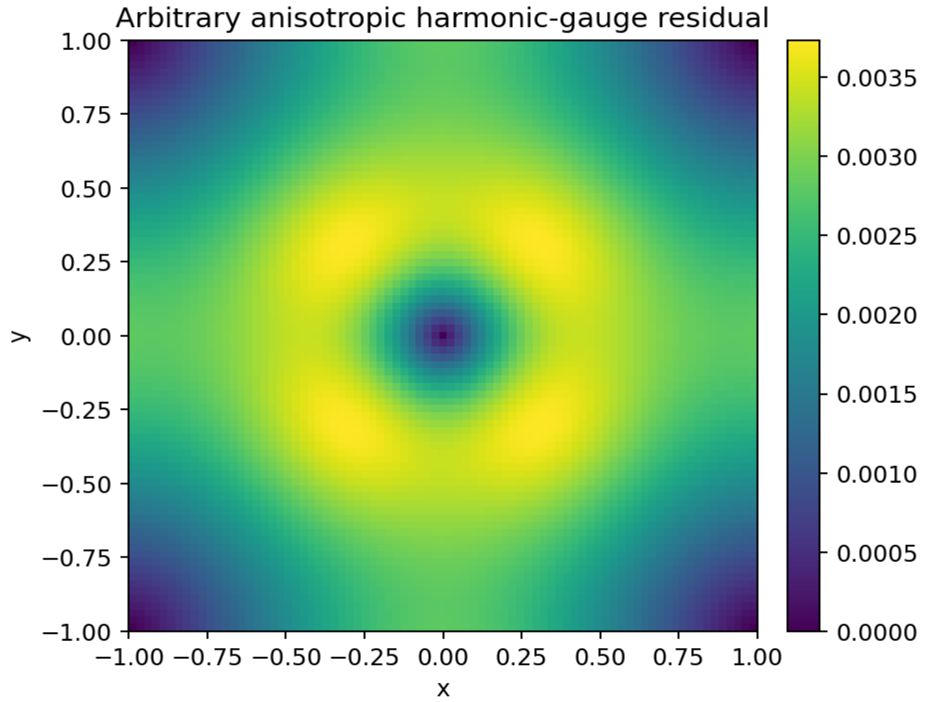


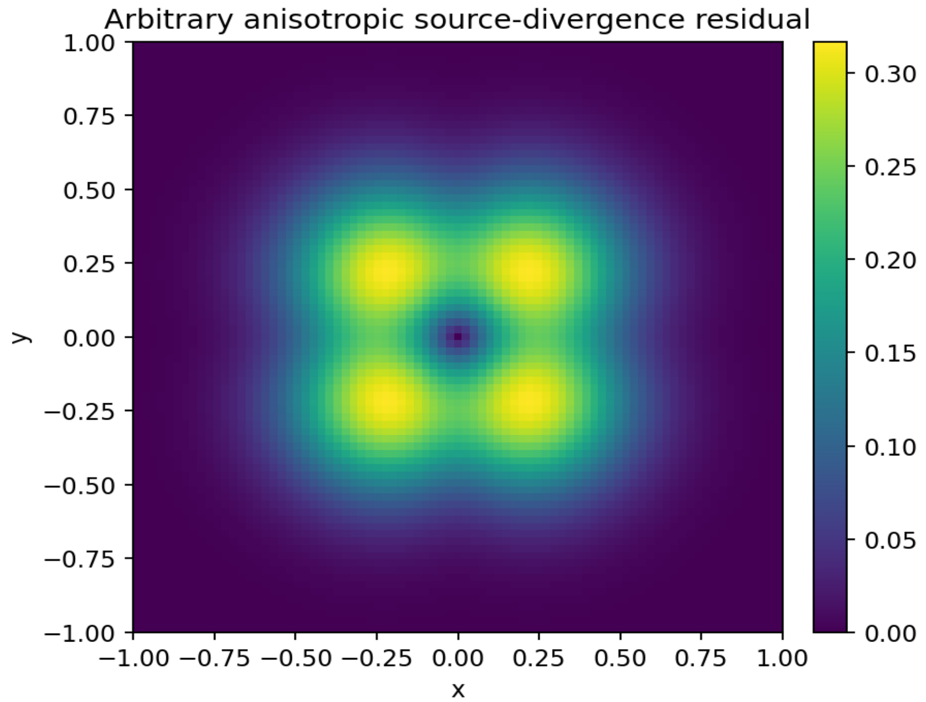


5. Gauge and source-conservation tests

A valid weak-field spin-2 source must be conserved. In this static 1+2D toy model, the flat source-divergence diagnostic is partial_i T_i_nu.







Source case	source div max	gauge max	trace residual	trace corr.	direct residual	Lorentz sig.
Dust conserved	0.000e+00	0.000e+00	5.534844e-03	0.999985	1.000000e+00	4761/4761
Arbitrary anisotropic	3.167e-01	3.732e-03	3.138810e-01	0.946899	4.614411e-01	4761/4761
Airy conserved stress	5.755e-03	8.968e-04	6.875743e-02	0.997662	7.411861e-01	4761/4761

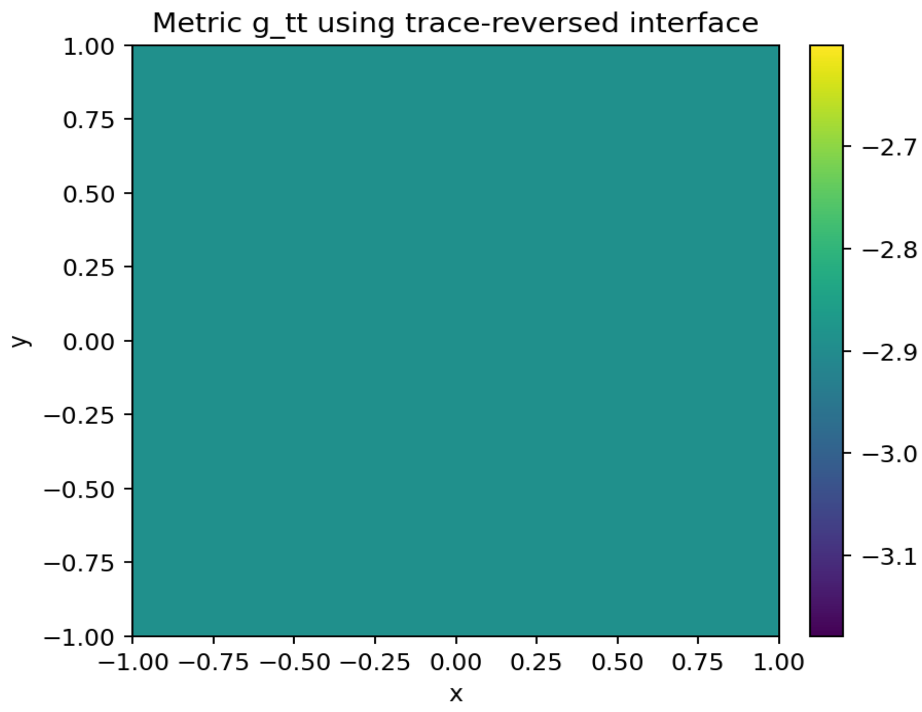
Interpretation: conservation and gauge compatibility are not optional. The arbitrary anisotropic source solves the component Poisson equations, but it violates the gauge/conservation structure and balance degrades.

6. Trace-reversed versus direct interface

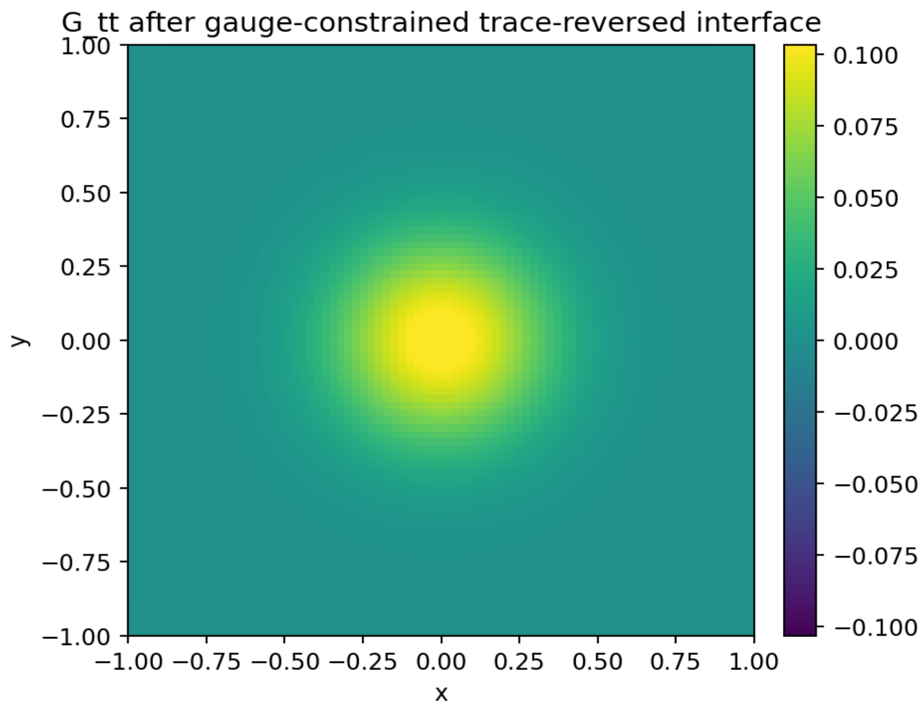
The same hbar solution was mapped two ways:

```
Direct:
h_mu_nu = hbar_mu_nu

Trace-reversed:
h_mu_nu = hbar_mu_nu - eta_mu_nu hbar_trace/(D-2)
```

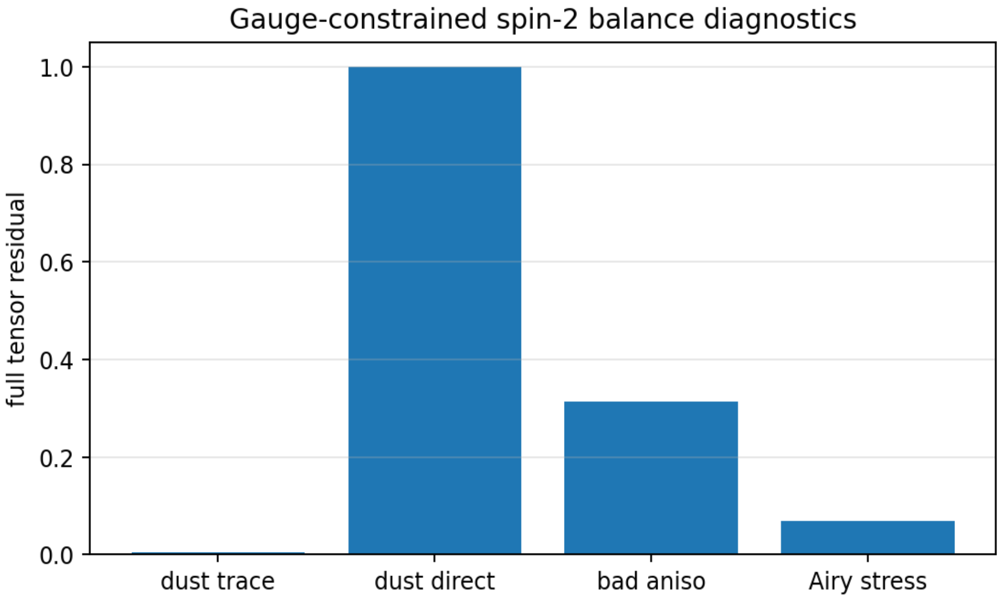
Result: direct mapping fails. Trace reversal remains the selected weak-field spin-2 interface.

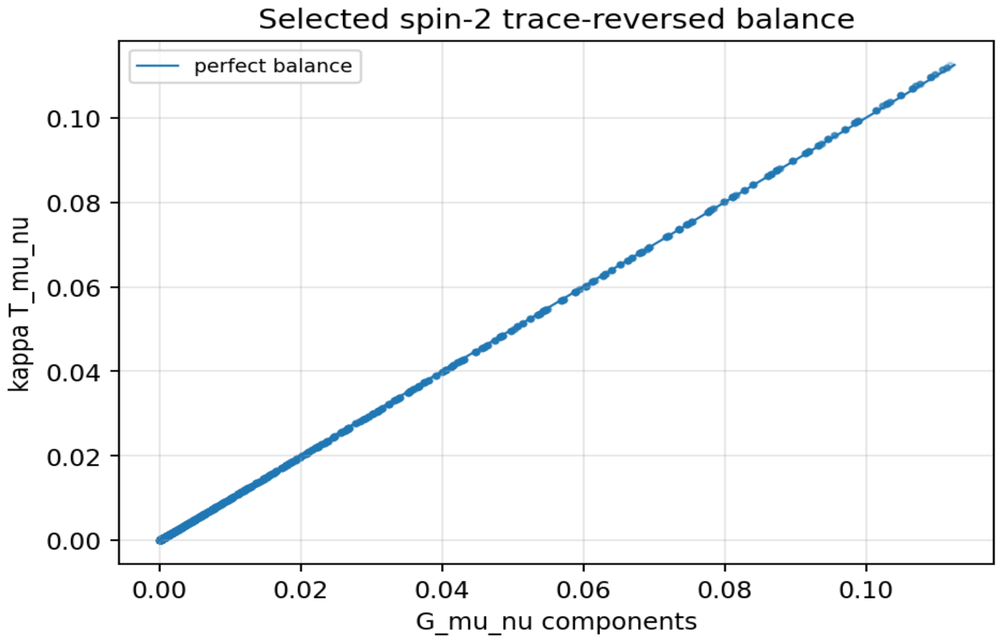


Metric g_tt using trace-reversed interface
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
y
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
x
−2.7
−2.8
−2.9
−3.0
−3.1



G_tt after gauge-constrained trace-reversed interface
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
y
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
x
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100



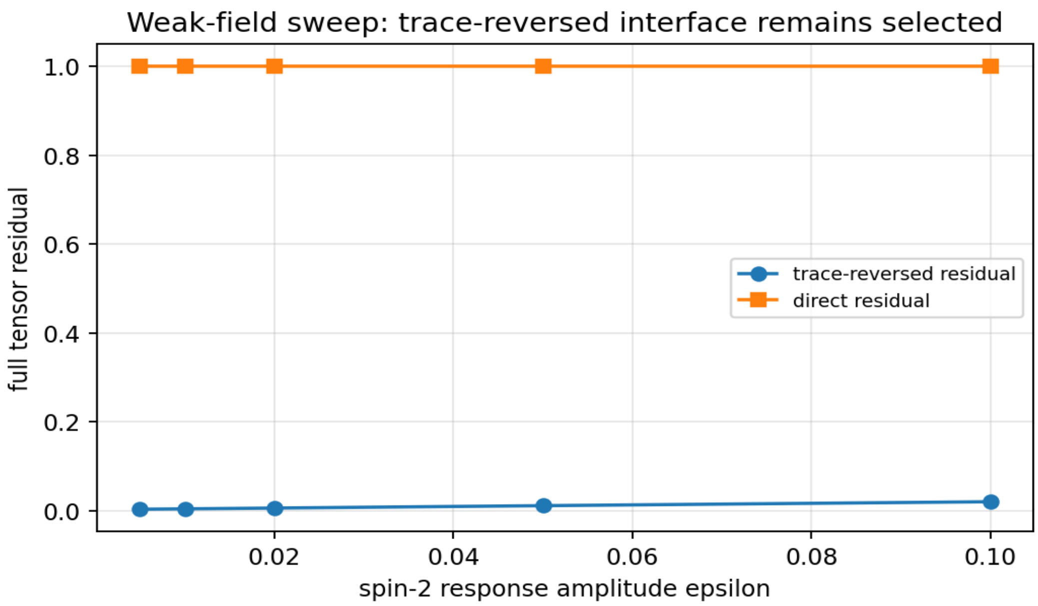


7. Component-by-component balance

Component	Residual	Correlation	Max |G|	Max |kT|
tt	5.534844e-03	0.999990	1.119311e-01	1.125261e-01
tx	0.000000e+00	1.000000	0.000000e+00	0.000000e+00
ty	0.000000e+00	1.000000	0.000000e+00	0.000000e+00
xx	4.354061e-02	1.000000	6.938894e-18	0.000000e+00
xy	1.000000e+00	0.000000	5.342850e-08	0.000000e+00
yy	4.356625e-02	1.000000	6.938894e-18	0.000000e+00

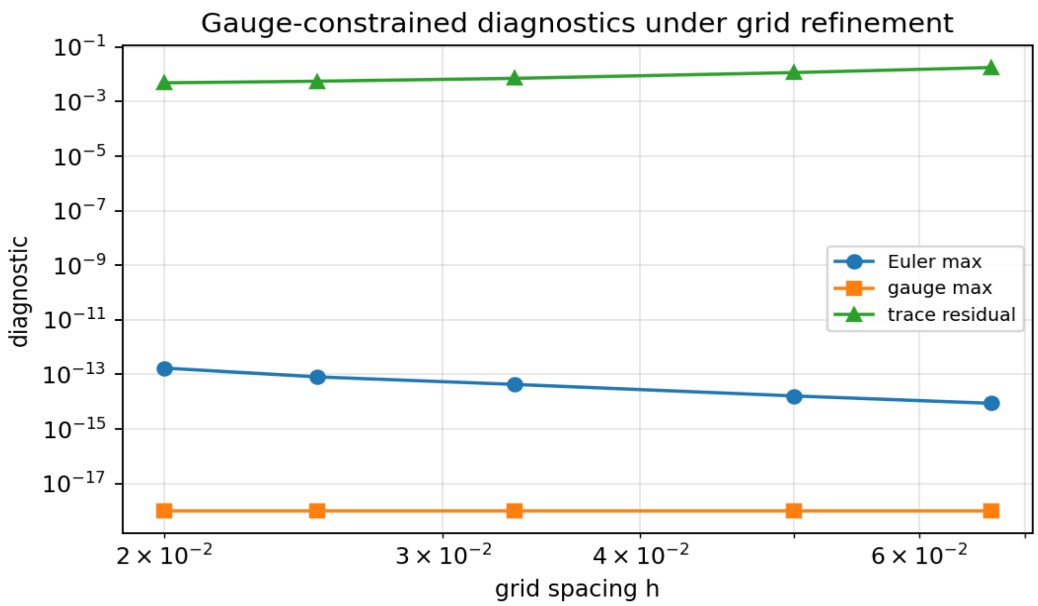
8. Weak-field amplitude sweep

The dust source was tested over several perturbation strengths. Trace reversal remains selected across the sweep, while the direct interface remains invalid.



epsilon	trace residual	trace corr.	direct residual	direct corr.	gauge max	Lorentz sig.
0.005	2.881502e-03	0.999996	1.000000e+00	-0.031531	0.000e+00	4761/4761
0.010	3.759728e-03	0.999993	1.000000e+00	-0.031516	0.000e+00	4761/4761
0.020	5.534844e-03	0.999985	1.000000e+00	-0.031486	0.000e+00	4761/4761
0.050	1.088528e-02	0.999942	1.000000e+00	-0.031397	0.000e+00	4761/4761
0.100	1.976698e-02	0.999808	1.000000e+00	-0.031247	0.000e+00	4761/4761

9. Grid-refinement behavior



N	h	Euler max	gauge max	trace residual	direct residual	Lorentz sig.
31	0.06667	8.632e-15	0.000e+00	1.763935e-02	1.000000e+00	529/529
41	0.05000	1.613e-14	0.000e+00	1.148196e-02	1.000000e+00	1089/1089
61	0.03333	4.249e-14	0.000e+00	7.078366e-03	1.000000e+00	2809/2809
81	0.02500	8.046e-14	0.000e+00	5.545654e-03	1.000000e+00	5041/5041
101	0.02000	1.696e-13	0.000e+00	4.834121e-03	1.000000e+00	7569/7569

10. Bianchi / conservation diagnostic

Diagnostic	Value
Dust action Euler residual max	8.046341e-14
Dust harmonic-gauge residual max	0.000000e+00

Diagnostic	Value
Dust source-divergence max	0.000000e+00
Dust covariant div(T) max	0.000000e+00
Dust geometry div(G) max	4.722339e-07
Dust geometry div(G) mean	1.654471e-07

11. Core numerical record

Diagnostic	Value
Grid	81 x 81
Interior points	4761
Lorentzian signatures	4761/4761
Source center	1.000000e+00
hbar_tt center	2.000000e-02
g_tt center, trace-reversed	-2.890000e+00
g_tt center, direct	-2.870000e+00
G_tt center, trace-reversed	1.119311e-01
G_tt center, direct	7.037457e-07
Ricci scalar center, trace-reversed	7.746094e-02

12. Interpretation inside the V4.2 framework

Toy Model 4J supports this limited ladder:

```
tensorial relational action
      + harmonic gauge constraint
      + conserved source
      -> trace-reversed spin-2 response variable hbar_mu_nu
      -> tensor Poisson equation
      -> metric perturbation through trace reversal
      -> weak-field curvature balance
```

It does not support this stronger ladder:

```
pure graph principles alone
      -> full nonlinear Einstein-Hilbert action
      -> physical stress-energy of real matter
      -> Newton's constant
      -> physical gravity / dark sector solution
```

13. What Toy Model 4J achieves

• It adds the missing weak-field gauge/spin-2 constraint to the 4I tensor action.

• It confirms that conserved sources are required for gauge-compatible spin-2 response.

• It confirms that the trace-reversed variable is the correct weak-field variable.

• It confirms that direct h=hbar mapping fails.

• It shows arbitrary anisotropic stress cannot be inserted without conservation/gauge control.

• It cleanly maps the toy model to the linearized Einstein-equation structure.

14. What Toy Model 4J does not achieve

• It does not derive the nonlinear Einstein-Hilbert action.

• It does not derive exact physical stress-energy.

• It does not derive Newton's gravitational constant.

• It does not derive full General Relativity.

• It does not solve dark matter or dark energy.

• It does not prove physical 3T.

15. Close-out statement for 4J

Toy Model 4J closes the weak-field spin-2 interface gap left by 4I. The trace-reversed response is no longer just a numerical preference; it is the natural variable of the harmonic-gauge spin-2 action. Conserved sources produce gauge-compatible response and strong weak-field balance, while non-conserved arbitrary anisotropic sources fail the gauge/balance test. The correct conclusion is structural compatibility with the linearized Einstein-equation layer, not a derivation of nonlinear gravity.

16. Recommended next chapter

The next logical move is Toy Model 4K: Nonlinear Self-Coupling and Einstein-Hilbert Bridge, or else a Gravity Track Close-Out Paper for 4A-4J before moving further.

1. Option A: close the gravity weak-field chapter now and produce a consolidated 4A-4J paper.

2. Option B: attempt nonlinear self-coupling, where the spin-2 field's own energy also becomes source.

3. Do not move to dark matter or dark energy until the weak-field chapter is consolidated.

4. Do not claim full GR until nonlinear self-coupling is handled.

17. Rebind prompt for continuation

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We are resuming after Toy Model 4J.

4I:
Tensor action gave component-wise tensor Poisson equations.
Trace-reversed interface worked; direct h=hbar failed.
But trace reversal was still selected by consistency.

4J:
Added weak-field spin-2 / harmonic-gauge constraint.

Gauge-constrained action:
A[hbar] =
integral [
  1/2 sum |grad hbar_mu_nu|^2
  - sum T_mu_nu hbar_mu_nu
  + (lambda_g/2) sum (partial^mu hbar_mu_nu)^2
] dA

Gauge condition:
partial^mu hbar_mu_nu = 0

Source condition:
partial_mu T^mu_nu = 0

Field equation in gauge sector:
nabla^2 hbar_mu_nu = -T_mu_nu

Metric reconstruction:
h_mu_nu = hbar_mu_nu - eta_mu_nu hbar_trace/(D-2)

Result:
Trace-reversed spin-2 interface is now structurally selected by gauge compatibility.
Direct h=hbar fails.
Non-conserved arbitrary anisotropic source violates gauge/balance.

Critical guardrail:
4J is linear weak-field structural compatibility.
It does not derive nonlinear Einstein-Hilbert action, physical GR, Newton's G,
dark matter, dark energy, or physical 3T.

Recommended next:
Either produce Gravity Track Close-Out Paper 4A-4J,
or move to Toy Model 4K - Nonlinear Self-Coupling and Einstein-Hilbert Bridge.
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